

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

MI-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mine Safety Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. What are the different approaches to safety risk assessment in mines? Explain Fault Tree Analysis, Failure mode and effects analysis, HAZOP with mining examples. 14
2. a) Classify the different kinds of accidents that take place in mines. Explain in brief. How would you minimise them? 8
b) State the features of safety management system and how will you minimise the risk in mines? 6
3. a) An accident occurred in a depillaring district of an underground coal mine due to 'air-blast' and one crew of a driller and a helper who were taking rest in the gallery were killed on the spot. As a mining engineer prepare a report on the accident occurred in a mine using the standard format. 10
b) Discuss the role of safety officer in mines. 4

MI-7005(2)-CBGS

PTO

[2]

4. a) Explain the purpose and procedure for investigation of accident. 7
b) Explain the duties of workmen's inspector as per The Mine Rules, 1955. 7
5. a) Explain assessment and evaluation of training schemes. 7
b) Write a note on "Safety Management Plan" as per the CMR, 2017. 7
6. a) Explain the appraisal of safety program. 7
b) What is the role and function of Directorate general of Mines Safety? 7
7. a) Explain how accident proneness is determined in mining operation. 3
b) Define Minor Injury, Reportable Injury and Serious Bodily Injury. 3
c) Write a note on "ergonomic controls" in mines with an example. 8
8. Write short notes on (any four) 14
 - a) Safety instruction
 - b) Job safety analysis
 - c) Personal Protective Equipment
 - d) Court of Enquiry
 - e) Cost of Accident

MI-7005(2)-CBGS